F3XP M.A.R.S.

A Level Physics Practice Paper 2

CANDIDATE NAME

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PHYSICS **9702/42**

Paper 4 A Level Structured Questions **August/September 2026**

2 hours

You must answer on the question paper.

No additional materials are needed.

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- You may use a calculator.
- You should show all your working and use appropriate units.

INFORMATION

- The total mark for this paper is 100.
- The number of marks for each question or part question is shown in brackets [ ].

This document has **24** pages. Any blank pages are indicated.

Data

acceleration of free fall	$g = 9.81\ \text{m s}^{-2}$
speed of light in free space	$c = 3.00 \times 10^{8}\ \text{m s}^{-1}$
elementary charge	$e = 1.60 \times 10^{-19}\ \text{C}$
unified atomic mass unit	$1\ \text{u} = 1.66 \times 10^{-27}\ \text{kg}$
rest mass of proton	$m_\text{p} = 1.67 \times 10^{-27}\ \text{kg}$
rest mass of electron	$m_\text{e} = 9.11 \times 10^{-31}\ \text{kg}$
Avogadro constant	$N_\text{A} = 6.02 \times 10^{23}\ \text{mol}^{-1}$
molar gas constant	$R = 8.31\ \text{J K}^{-1}\ \text{mol}^{-1}$
Boltzmann constant	$k = 1.38 \times 10^{-23}\ \text{J K}^{-1}$
gravitational constant	$G = 6.67 \times 10^{-11}\ \text{N m}^2\ \text{kg}^{-2}$
permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ \text{F m}^{-1}$ $\left(\frac{1}{4\pi\varepsilon_0} = 8.99 \times 10^{9}\ \text{m F}^{-1}\right)$
Planck constant	$h = 6.63 \times 10^{-34}\ \text{J s}$
Stefan–Boltzmann constant	$\sigma = 5.67 \times 10^{-8}\ \text{W m}^{-2}\ \text{K}^{-4}$

Formulae

uniformly accelerated motion	$s = ut + \frac{1}{2}at^2$ $v^2 = u^2 + 2as$
hydrostatic pressure	$\Delta p = \rho g \Delta h$
upthrust	$F = \rho g V$
Doppler effect for sound waves	$f_\text{o} = \frac{f_\text{s} v}{v \pm v_\text{s}}$
electric current	$I = Anvq$
resistors in series	$R = R_1 + R_2 + ...$
resistors in parallel	$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + ...$

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gravitational potential	$\phi = -\frac{GM}{r}$
gravitational potential energy	$E_P = -\frac{GMm}{r}$
pressure of an ideal gas	$p = \frac{1}{3}\frac{Nm}{V}\langle c^2 \rangle$
simple harmonic motion	$a = -\omega^2 x$
velocity of particle in s.h.m.	$v = v_0 \cos \omega t$ $v = \pm \omega \sqrt{(x_0^2 - x^2)}$
electric potential	$V = \frac{Q}{4\pi\varepsilon_0 r}$
electrical potential energy	$E_P = \frac{Qq}{4\pi\varepsilon_0 r}$
capacitors in series	$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + ...$
capacitors in parallel	$C = C_1 + C_2 + ...$
discharge of a capacitor	$x = x_0 e^{-\frac{t}{RC}}$
Hall voltage	$V_H = \frac{BI}{ntq}$
alternating current/voltage	$x = x_0 \sin \omega t$
radioactive decay	$x = x_0 e^{-\lambda t}$
decay constant	$\lambda = \frac{0.693}{t_{½}}$
intensity reflection coefficient	$\frac{I_R}{I_0} = \frac{(Z_1 - Z_2)^2}{(Z_1 + Z_2)^2}$
Stefan–Boltzmann law	$L = 4\pi\sigma r^2 T^4$
Doppler redshift	$\frac{\Delta\lambda}{\lambda} \approx \frac{\Delta f}{f} \approx \frac{v}{c}$

1 A small ball of mass *m* is attached to one end of a light inextensible string. The other end of the string is attached to a fixed point O. The ball moves in a vertical circle of radius *r* with centre O, as shown in Fig. 1.1.

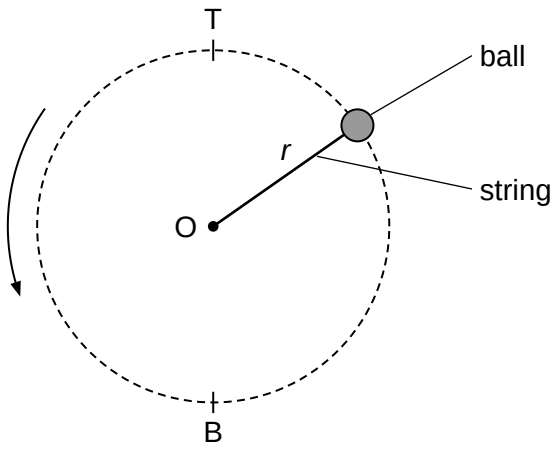


Fig. 1.1

Point T is at the top of the circle and point B is at the bottom of the circle. Air resistance is negligible.

(a) Show that the minimum speed of the ball at T for which the string remains taut is $\sqrt{(gr)}$, where *g* is the acceleration of free fall.

[2]

(b) The ball passes through T with speed *v*. A student suggests that the tension in the string when the ball is at B is greater than the tension when the ball is at T by $2mg$, because the weight of the ball acts towards O at T but away from O at B.

Determine, in terms of *m* and *g* only, the difference between the tension in the string at B and the tension in the string at T. Hence state whether the student is correct.

difference in tension = ... [3]

(c) The ball has mass 0.15 kg and the radius *r* of the circle is 0.60 m. The string breaks if the tension in it exceeds 12.0 N.

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Determine the range of values of the speed of the ball at T for which the ball moves in a complete vertical circle without the string breaking.

minimum speed = ... m s^{-1}

maximum speed = ... m s^{-1} [3]

(d) The ball and string are replaced by a small block that slides without friction along a track. The track includes a vertical circular loop of radius *r*, as shown in Fig. 1.2.

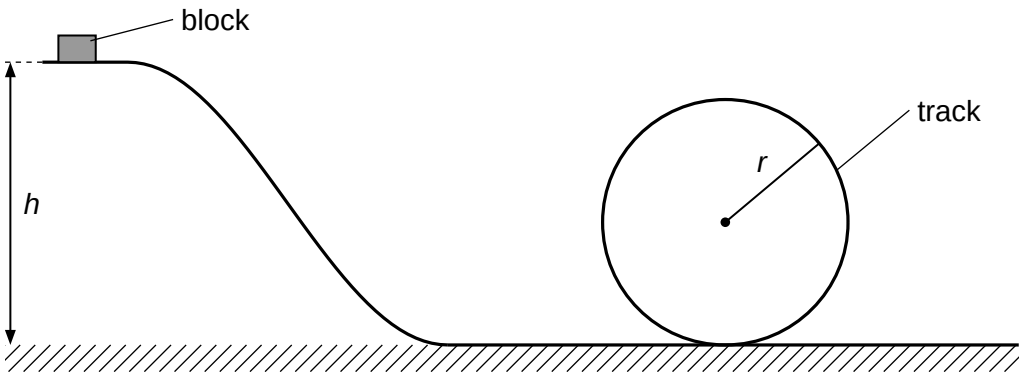


Fig. 1.2

The block is released from rest at a height *h* above the lowest point of the loop. A student suggests that the block will just reach the top of the loop if $h = 2r$.

Explain why the student is wrong and determine, in terms of *r*, the minimum value of *h* for which the block moves all the way round the loop.

minimum *h* = ... [2]

[Total: 10]

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2 A planet may be considered to be an isolated uniform sphere of radius 4.0×10^6 m.

Fig. 2.1 shows the variation of the gravitational potential ϕ with distance r from the centre of the planet, for points at and above the surface of the planet.

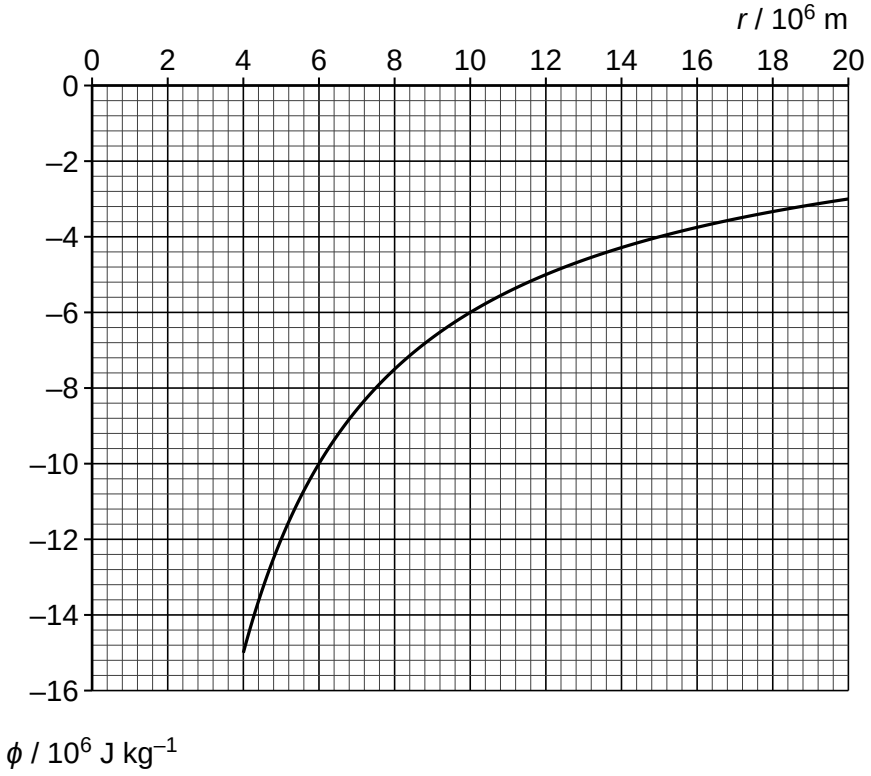


Fig. 2.1

(a) Explain why the gravitational potential at every point in the field of the planet is negative.

.....

.....

..... [2]

(b) Use Fig. 2.1 to determine the gravitational field strength at a distance of 8.0×10^6 m from the centre of the planet. Show your working on Fig. 2.1.

field strength = ... N kg^{-1} [2]

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(c) The planet rotates about its axis once every 12.0 hours. A satellite of mass 250 kg is to be placed in an equatorial orbit so that it remains above the same point on the surface of the planet, as illustrated in Fig. 2.2.

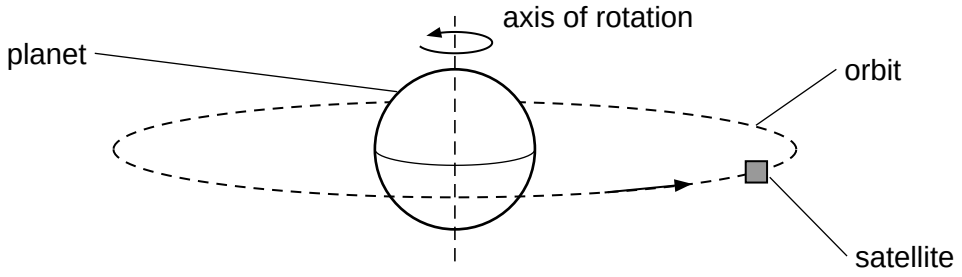


Fig. 2.2 (not to scale)

(i) Use Fig. 2.1 to show that the radius of the orbit is 1.4×10^7 m.

[3]

(ii) Determine the minimum energy needed to move the satellite from rest on the surface of the planet into this orbit. Ignore the rotation of the planet.

energy = ... J [3]

(iii) The satellite is launched from a point on the equator of the planet. Suggest why the energy that must be supplied is less than your answer to (c)(ii) when the rotation of the planet is taken into account.

.....

..... [1]

[Total: 11]

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3 **(a)** Ice cubes of total mass 150 g at a temperature of –18.0 °C are put into 200 g of water at a temperature of 30.0 °C in a vacuum flask. The flask has negligible heat capacity and no thermal energy is transferred between the contents of the flask and the surroundings.

The specific heat capacity of ice is 2.10 J g^{-1} $°C^{-1}$ and the specific heat capacity of water is 4.18 J g^{-1} $°C^{-1}$. The specific latent heat of fusion of ice is 334 J g^{-1}.

(i) Determine the temperature of the contents of the flask when thermal equilibrium is reached.

temperature = ... °C [3]

(ii) Determine the mass of liquid water in the flask when thermal equilibrium is reached.

mass = ... g [2]

(b) Fig. 3.1 shows some air sealed in a syringe by a piston.

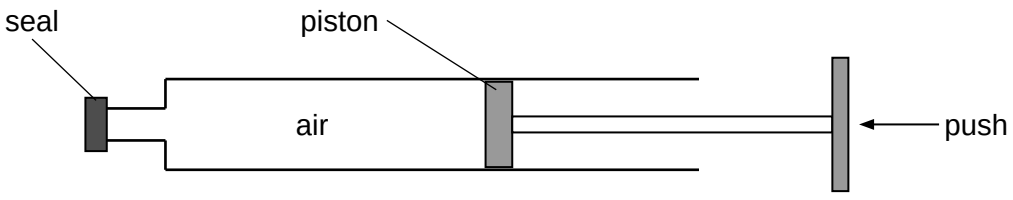


Fig. 3.1

The air has an initial volume of 150 cm^3 and an initial pressure of 100 kPa. The piston is pushed in so rapidly that no thermal energy is transferred between the air and its surroundings. The volume of the air decreases to 75 cm^3. The work done on the air during this compression is 12 J. The air may be assumed to be an ideal gas.

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(i) A student states that the temperature of the air does not change during the compression, because no thermal energy is supplied to the air. Use the first law of thermodynamics to explain why the student is wrong.

.....

.....

..... [2]

(ii) Fig. 3.2 shows the variation with volume V of the pressure p of the air during the rapid compression.

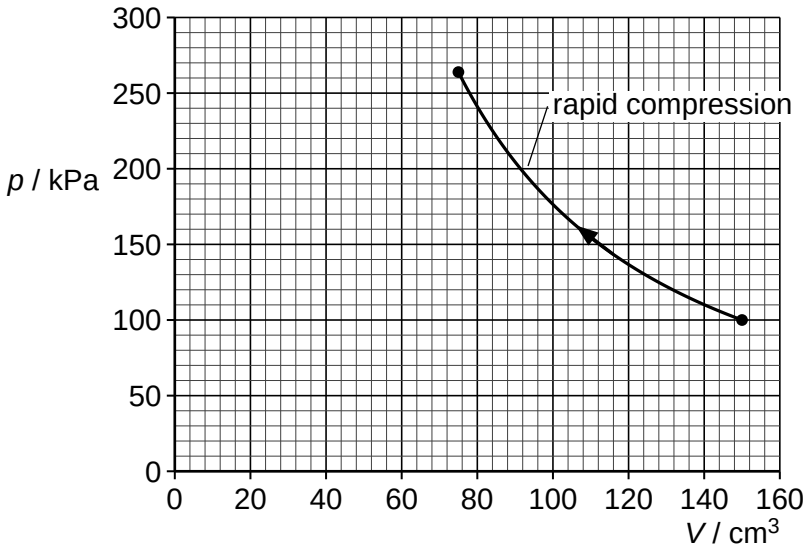


Fig. 3.2

The air is returned to its initial state. It is then compressed slowly from 150 cm^3 to 75 cm^3 so that its temperature remains constant. On Fig. 3.2, sketch the variation of p with V for this slow compression. [2]

(iii) Use Fig. 3.2 to explain whether the thermal energy transferred during the slow compression is greater than, equal to or less than 12 J. State the direction of this transfer of thermal energy.

.....

.....

..... [2]

[Total: 11]

4 **(a)** The pressure p of an ideal gas is given by

$$p = \frac{1}{3}\frac{Nm}{V}\langle c^2 \rangle$$

where the symbols have their usual meanings. Use this equation to show that the mean translational kinetic energy of a molecule of an ideal gas is (3/2)kT, where k is the Boltzmann constant and T is the thermodynamic temperature of the gas.

[2]

(b) Air consists mainly of nitrogen molecules, each of mass 28.0 u, and oxygen molecules, each of mass 32.0 u.

A student states that, in a sample of air, the oxygen molecules have a greater mean kinetic energy than the nitrogen molecules because each oxygen molecule has a greater mass.

(i) Explain why the student is wrong.

.....

..... [1]

(ii) Calculate, for the sample of air, the ratio

$$\frac{\text{r.m.s. speed of the nitrogen molecules}}{\text{r.m.s. speed of the oxygen molecules}}.$$

ratio = ... [1]

(c) The envelope of a hot-air balloon has a fixed volume of 2800 m^3. It is open at its base, so that the pressure of the air inside it is always equal to the atmospheric pressure of 1.01×10^5 Pa. Initially, the air inside the envelope is at the same temperature as the air outside, 15 °C. A burner then heats the air inside the envelope to 100 °C. The molar mass of air is 0.029 kg mol^{-1}.

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Fig. 4.1 shows the balloon.

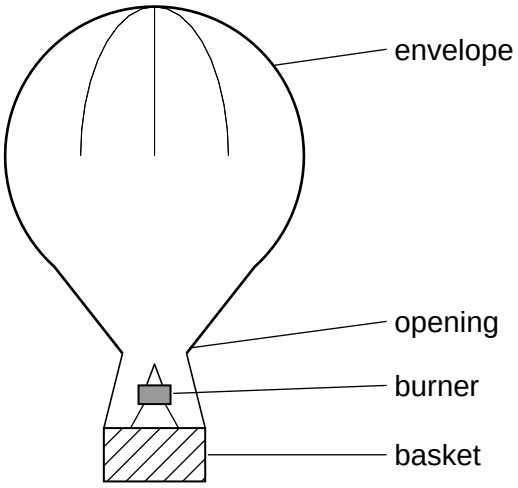


Fig. 4.1

(i) A student states that the pressure of the air inside the envelope increases as it is heated, because p / T is constant for an ideal gas. Explain why the student is wrong.

.....

..... [1]

(ii) Determine the mass of air that leaves the envelope as the air inside it is heated from 15 °C to 100 °C.

mass = ... kg [3]

(iii) Use your answer to (c)(ii) to deduce the maximum total mass of the envelope, the basket, the burner and the passengers that the balloon can lift when the air inside the envelope is at 100 °C. Explain your reasoning.

.....

.....

maximum mass = ... kg [1]

[Total: 9]

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5 Fig. 5.1 is a view from above of a peg P fixed to the edge of a horizontal turntable of radius *r*.

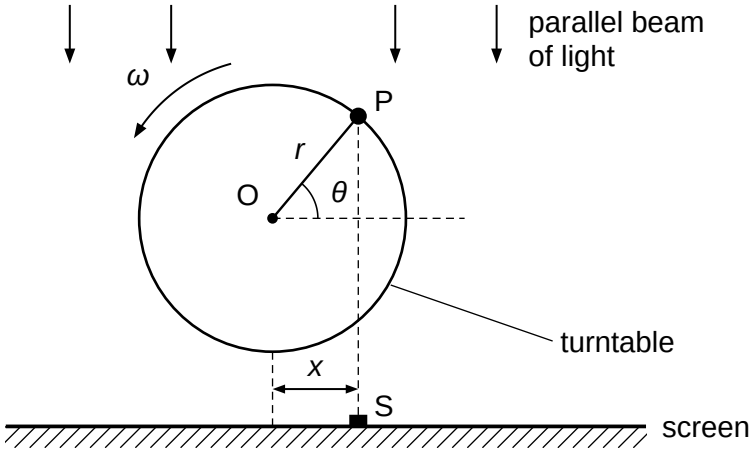


Fig. 5.1 (view from above)

The turntable rotates about its centre O with constant angular speed *ω*. A horizontal parallel beam of light casts a shadow S of the peg on a vertical screen. At the instant shown, the line OP makes an angle *θ* with a line through O that is parallel to the screen. The displacement of S from the point on the screen directly opposite O is *x*.

(a) By considering the acceleration of P, explain why S moves with simple harmonic motion.

.....

.....

.....

..... [3]

(b) The angular speed of the turntable is doubled. State the factor by which the maximum acceleration of S increases.

factor = ... [1]

(c) A mass on a spring is made to oscillate by a periodic driving force of frequency *f*. Fig. 5.2 shows the variation with *f* of the amplitude *A* of the oscillations when the damping is light.

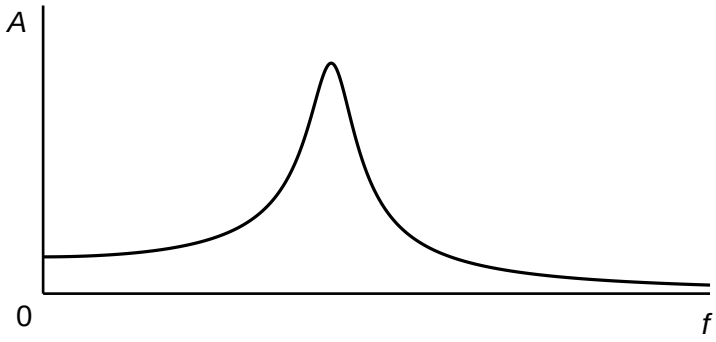


Fig. 5.2

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On Fig. 5.2, sketch a line to show the variation with f of A when the damping is increased.

[2]

(d) Two oscillators X and Y have the same frequency. Fig. 5.3 shows the variation with time t of the displacement x of X and the variation with t of the velocity v of Y.

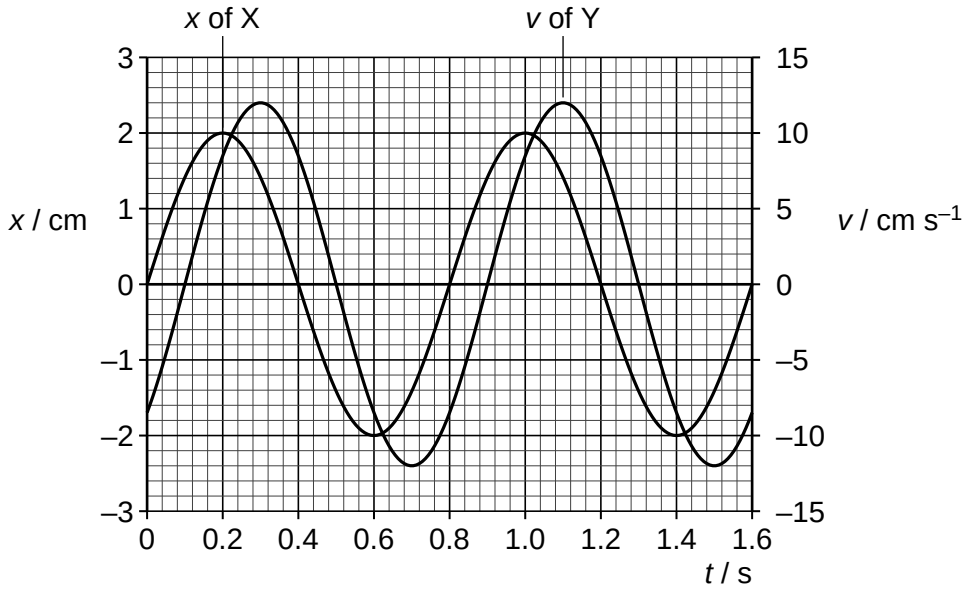


Fig. 5.3

(i) Determine the phase difference between the displacement of X and the displacement of Y.

phase difference = ... ° [2]

(ii) Determine the amplitude of the oscillations of Y.

amplitude = ... cm [2]

[Total: 10]

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6 **(a)** Two point charges A and B are separated by a distance of 0.30 m in a vacuum, as shown in Fig. 6.1.

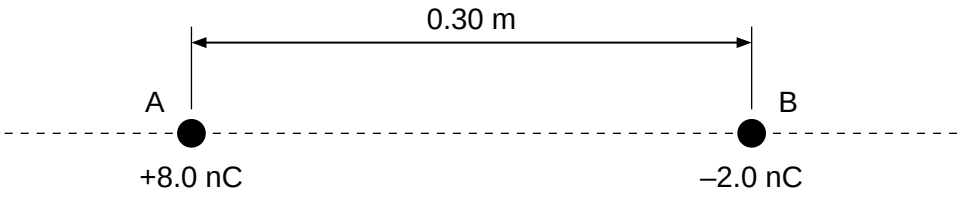


Fig. 6.1

The charge of A is +8.0 nC and the charge of B is –2.0 nC.

(i) Determine the distance from A of the point on the line through A and B at which the electric field strength is zero.

distance = ... m [3]

(ii) Calculate the electric potential at the point in (a)(i).

potential = ... V [2]

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(b) Electrons are accelerated from rest through a potential difference (p.d.) of 1500 V. They then enter the region between two horizontal metal plates, midway between the plates, as shown in Fig. 6.2.

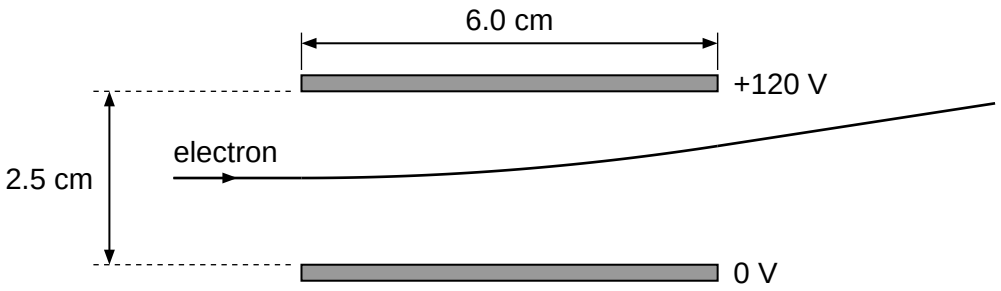


Fig. 6.2 (not to scale)

The plates have length 6.0 cm and are separated by 2.5 cm. The p.d. between the plates is 120 V. The electrons enter the region between the plates with speed 2.30×10^7 m s^{-1} in a direction parallel to the plates. Assume that the electric field is uniform between the plates and zero outside them.

(i) Determine the vertical distance moved by an electron while it is between the plates.

distance = ... m [3]

(ii) The electrons are replaced by protons that are accelerated from rest through the same p.d. of 1500 V and then enter the region between the plates at the same point.

A student suggests that the protons will be deflected much less than the electrons, because the mass of a proton is much greater than the mass of an electron.

Explain why the student is wrong, and describe the deflection of the protons.

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..... [2]

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7 **(a)** Define capacitance.

.....

..... [1]

(b) Fig. 7.1 shows a circuit containing two identical capacitors X and Y, each of capacitance *C*, a resistor of resistance *R*, a two-way switch S and a battery of electromotive force (e.m.f.) 12.0 V and negligible internal resistance.

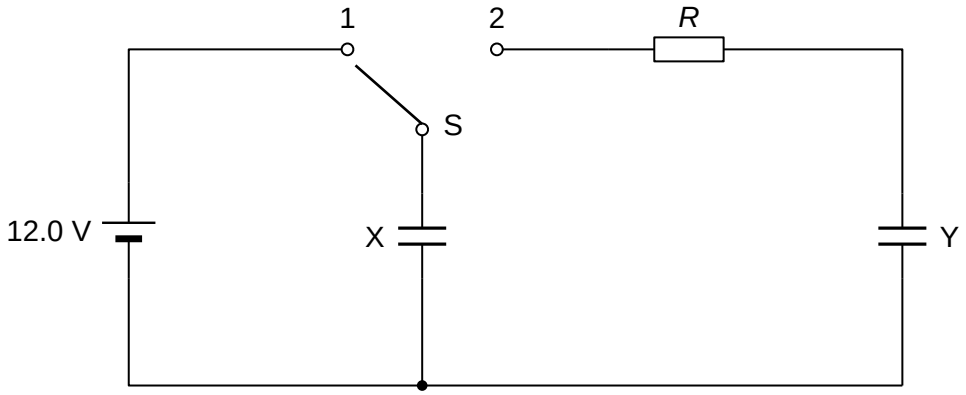


Fig. 7.1

Capacitor Y is initially uncharged. Switch S is first moved to position 1, so that X becomes fully charged. S is then moved to position 2. Charge flows until the potential difference (p.d.) across X is equal to the p.d. across Y.

(i) Show that the final p.d. across each capacitor is 6.0 V.

[1]

(ii) Show that half of the energy that was stored in X before S was moved to position 2 is dissipated in the resistor.

[2]

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(iii) A student suggests that less energy would be dissipated if a resistor of much smaller resistance were used. Explain why the student is wrong.

.....

.....

..... [2]

(c) After the current has fallen to zero, X and Y are disconnected from the rest of the circuit, while still connected to each other. They are then discharged together through a resistor of resistance 10.0 kΩ.

Fig. 7.2 shows the variation with time *t* of ln (*V* / V), where *V* is the p.d. across X during this discharge.

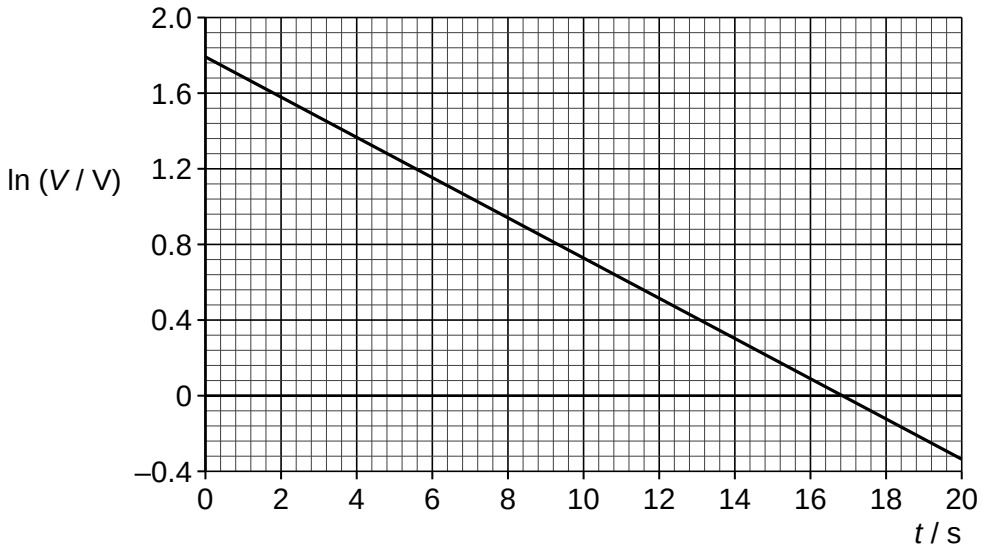


Fig. 7.2

Use Fig. 7.2 to determine the capacitance *C* of X.

C = ... μF [3]

[Total: 9]

8 Fig. 8.1 shows a flat rectangular coil in a uniform magnetic field between the poles of a magnet.

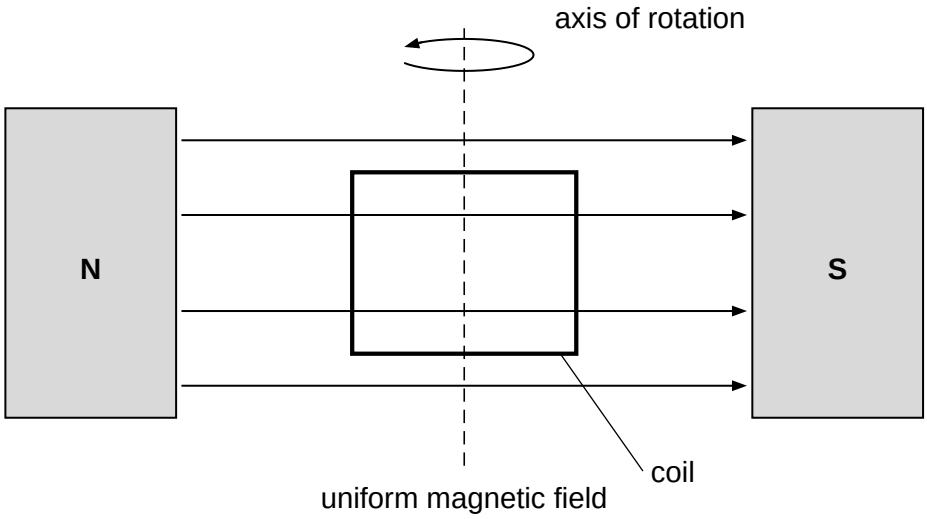


Fig. 8.1

The coil rotates with constant angular speed about an axis that is perpendicular to the magnetic field. At the instant shown, the plane of the coil is parallel to the magnetic field.

(a) State Faraday's law of electromagnetic induction.

.....

.....

..... [2]

(b) A student states that the electromotive force (e.m.f.) induced in the coil is zero at the instant shown in Fig. 8.1, because no magnetic flux passes through the coil at this instant.

Explain why the student is wrong.

.....

.....

..... [2]

(c) The ends of the coil are connected to a resistor of resistance 8.0 Ω. The e.m.f. E, in V, induced in the coil varies with time t, in s, according to the equation

$$E = 25 \sin (120\pi t).$$

The coil has negligible resistance.

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(i) Calculate the mean power dissipated in the resistor.

mean power = ... W [2]

(ii) The angular speed of the coil is doubled. Determine the new mean power dissipated in the resistor.

mean power = ... W [2]

(d) The coil is connected to the resistor by two long straight leads that are parallel to each other and close together, as shown in Fig. 8.2.

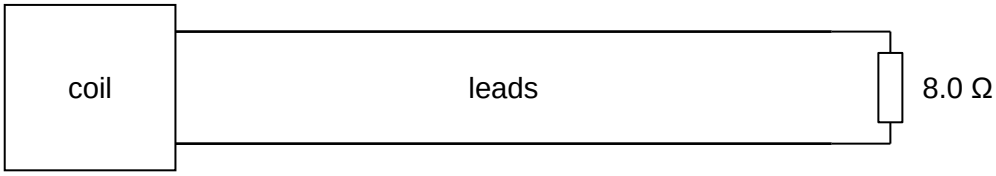


Fig. 8.2

(i) Explain why the magnetic force between the two leads is always repulsive, even though the current in the leads is alternating.

.....

..... [1]

(ii) For the e.m.f. in (c), state the frequency with which the magnitude of the force between the leads varies.

frequency = ... Hz [1]

[Total: 10]

9 X-rays are produced when electrons that have been accelerated from rest through a potential difference (p.d.) strike a molybdenum target.

Fig. 9.1 shows the spectrum of the X-rays produced when the accelerating p.d. is 28.0 kV.

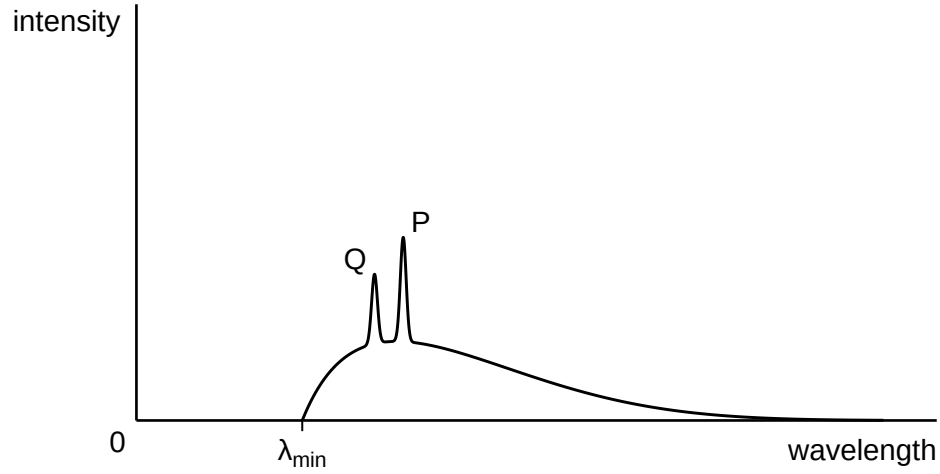


Fig. 9.1

The spectrum consists of a continuous range of wavelengths, with a minimum wavelength λ_{min}, together with two sharp peaks P and Q.

(a) Explain why there is a minimum wavelength.

.....

..... [1]

(b) Calculate λ_{min}.

λ_{min} = ... m [2]

(c) The current in the X-ray tube is doubled. The accelerating p.d. is unchanged.

On Fig. 9.1, sketch the spectrum of the X-rays that are now produced. [2]

(d) Each sharp peak is produced when an incident electron removes an electron from the lowest energy level of a molybdenum atom in the target. An electron from a higher energy level then moves into the vacancy, emitting a photon.

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Fig. 9.2 shows some of the energy levels of a molybdenum atom.

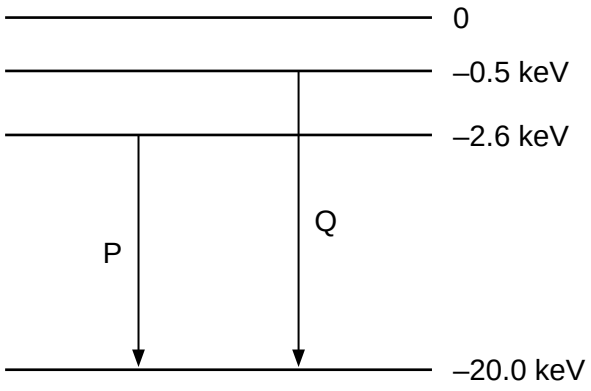


Fig. 9.2 (not to scale)

The photons in peak P have energy 17.4 keV. A student suggests that peak P would still be produced if the accelerating p.d. were reduced to 18.0 kV, because each electron would then have enough energy to produce a photon of energy 17.4 keV.

Use Fig. 9.2 to explain why the student is wrong.

.....

.....

..... [2]

(e) A different X-ray tube is used to produce an image of a limb. For the X-ray photons used, the half-value thickness (the thickness of material that halves the intensity of a parallel beam) is 3.5 cm for soft tissue and 1.3 cm for bone.

Along path A through the limb, a parallel beam of X-rays passes through 7.0 cm of soft tissue only. Along path B, the beam passes through 2.0 cm of bone and 5.0 cm of soft tissue.

Determine the ratio

$$\frac{\text{intensity of X-rays emerging along path B}}{\text{intensity of X-rays emerging along path A}}.$$

ratio = ... [3]

[Total: 10]

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10 **(a)** Fig. 10.1 shows the variation with nucleon number A of the binding energy per nucleon E_B of nuclei.

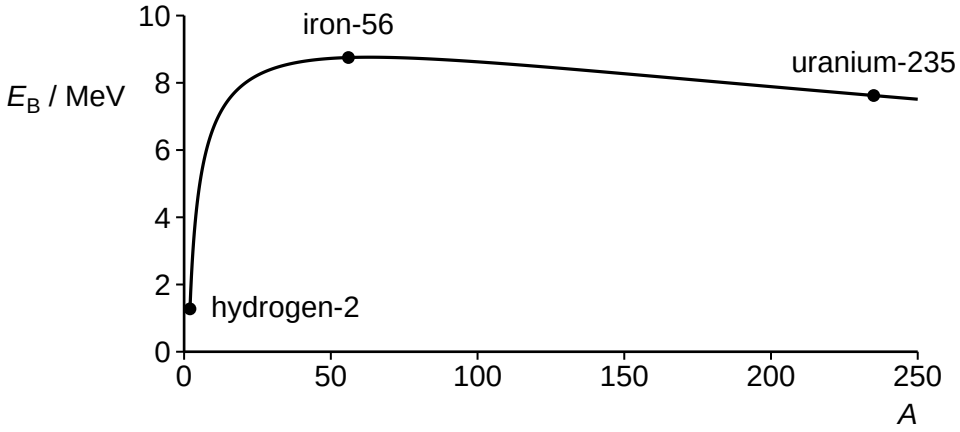


Fig. 10.1

A student suggests that, because energy is released when a nucleus of uranium-235 undergoes fission, energy must be absorbed when two nuclei of hydrogen-2 fuse, since fusion is the reverse of fission. Use Fig. 10.1 to explain why the student is wrong.

.....

.....

..... [2]

(b) A sample of a radioactive isotope is placed near a detector. Fig. 10.2 shows the variation with time t of the count rate C recorded by the detector. The recorded count rate includes a background count rate of 30 min^{-1}.

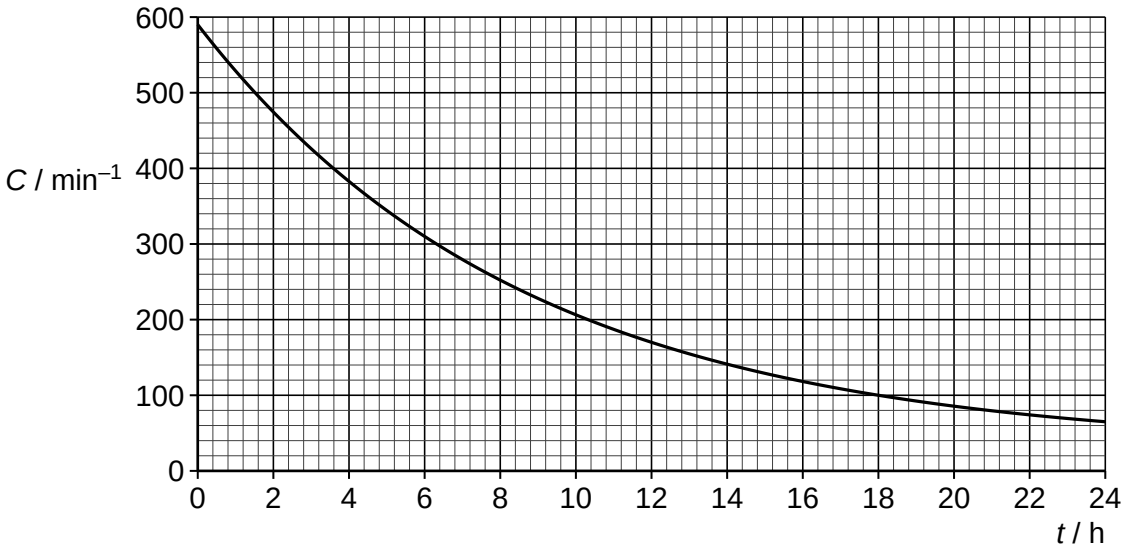


Fig. 10.2

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Use Fig. 10.2 to determine the half-life of the isotope.

half-life = ... h [3]

(c) Type Ia supernovae are used as standard candles.

(i) State what is meant by a standard candle.

.....

..... [1]

(ii) A type Ia supernova in a distant galaxy has a peak luminosity of 2.0×10^{36} W. The maximum radiant flux intensity received at the Earth from the supernova is 1.5×10^{-15} W m^{-2}. Determine the distance of the galaxy from the Earth.

distance = ... m [2]

(iii) A spectral line that has a wavelength of 656 nm when measured in a laboratory is observed in the light from the galaxy to have a wavelength of 708 nm. Use your answer to (c)(ii) to determine a value for the Hubble constant.

Hubble constant = ... s^{-1} [2]

[Total: 10]

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